

The Hidden Complexity of Mental Health: Multi-Entropy Analysis of Response Patterns in Depression Severity Assessment

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Abstract—The Patient Health Questionnaire-9 (PHQ-9) is generally used to determine the level of depression, and clinical decisions are usually made based on the overall score. Nevertheless, there can be significant variation in the response pattern of individuals who have the same score, indicating that there are variations in the organization of symptoms. To overcome this drawback, this paper presents an entropy-based computation system that describes PHQ-9 responses in a way that is not limited by a score measure. Responses to individual items are fused with information-theoretic quantities, such as Shannon entropy, Sample entropy, Permutation entropy, and Multiscale entropy, to form a more detailed feature space to classify depression severity. The proposed framework significantly increased the accuracy of the sum-score baseline in fivefold stratified cross-validation of the existing data, with the Random Forest adding 18.0 percentage points to the accuracy of 81.2 percent, reaching 99.2 percent. Statistical analysis also shows that the relationship between response entropy and the severity of depression is not linear, where entropy is at its highest when depression is moderate and reduced with severe depression. A potentially useful screening signal in this data set was observed to be an entropy threshold (0.439). These results suggest the complexity of the response patterns as a concomitant measure of digital depression.

Index Terms—Depression classification, Digital biomarkers, Mental health assessment, PHQ-9, Response pattern analysis

I. INTRODUCTION

Being a common psychological condition, depression manifests itself through constant sadness, loss of impetus, intellectual inability, and diminished functional and daily performance. It is a disease that is experienced by hundreds of millions of people around the world and is one of the main causes of disability and poor quality of life in this world [1]. Although clinically predominant, depression is also a heterogeneous disorder, i.e. people diagnosed with the same severity level can display and report symptoms in significantly different ways.

Clinical evaluation of the extent of depression is often done using standardized self-report scales such as the Patient Health Questionnaire-9 (PHQ-9) in which the severity of symptoms is measured by summation of scores on nine diagnostic questions [2]. Although this score-based model offers an effective and proven screening system, it is implicitly assumed that people with the same total scores have the same psychological state [3], [4]. However, in reality, patients can use radically different response patterns, with varying distributions of symptoms and underlying mental disorders, to achieve the same cumulative score. Additionally, significant information that has been embedded in individual response patterns can be missed when measurement is made based on summative scores.

Recent developments in computational psychiatry and digital mental health studies have highlighted the importance of discovering objective and data-driven biomarkers [5] that have the potential to provide behavioral complexity beyond classical linear measures. Entropy-based analysis, which is used in information-theoretic methods, provides a principled way to measure the variability, irregularity, and structural organization of behavioral responses. This is possible with measures which allow patterns of symptom expression to be characterized as opposed to concentrating solely upon symptom intensity.

To address these limitations, this paper proposes a multi-entropy framework for exploring the latent complexity of PHQ-9 response behavior. Multiple entropy measures are integrated with machine learning models to examine whether response organization provides information beyond conventional scoring for depression severity classification. In addition, statistical validation and unsupervised clustering are used to reveal non-linear relationships and latent response subtypes across clinically defined severity groups. Rather than replacing PHQ-9 scoring, the proposed framework complements the

conventional assessment by modeling how symptoms are expressed, supporting a more nuanced and data-driven approach to the evaluation of digital depression.

II. LITERATURE REVIEW

Recent computational psychiatry and digital mental health have focused on objective and data-driven biomarkers to measure the complexity of behavior that is not well represented by traditional linear measures of behavioral complexity [5]. Although standardized are still clinically valuable, sum-score based measurement can fail to capture key individual variations in symptom configuration [6], [7]. Earlier research has thus started to study psychometric properties in addition to total scores in order to determine the latent structure that traditional linear methods might not reveal [8].

Non-linear dynamics and information theory have become an effective method of explaining depressive disorders. Behavioral patterns have been analyzed using entropy measures to demonstrate the complexity of human reactions [9], [10]. In addition, we have techniques such as multiscale entropy or careful consideration of Likert responses that point to the fact that the disordered nature of the responses provides a more in-depth explanation of cognitive stability than the severity of the response [11].

Machine Learning (ML) methods have greatly contributed to the development of mental health research, providing more precise and plausible diagnostic results. These models can discover hidden patterns of behavior that are usually not apparent with traditional statistical techniques [12]. A combination of these computational methods can enhance traditional psychiatric testing by providing more specific data-driven support for clinical decision making [13]. Entropy-based analysis, as one of the information-theoretic approaches, provides a principled framework of how much variability, irregularity, and structural organization there is in behavioral responses.

III. METHODOLOGY

This study follows a structured and interpretable computational workflow, as illustrated in Fig. 1.

The objective is to transform raw PHQ-9 responses into clinically relevant machine-learning features using entropy theory. The pipeline consists of data collection and preprocessing, entropy computation, statistical validation, predictive modeling, and threshold optimization [1], [2].

A. Study Design and Dataset

We analyze 2,039 completed PHQ-9 questionnaires collected from university students from multiple institutions through an anonymized survey protocol [1], [14]. The Patient Health Questionnaire-9 (PHQ-9) is a standardized self-report instrument designed to assess the frequency of nine core depressive symptoms in the preceding two weeks. The items contained in the questionnaire are as follows:

- **PHQ-1:** Minimal interest / pleasure in doing things.
- **PHQ-2:** Feeling down, depressed, or hopeless.
- **PHQ-3:** Falling trouble, staying or sleeping too small.

- **PHQ-4:** Feeling fatigued or feeling low energy.
- **PHQ-5:** Not eating or overeating.
- **PHQ-6:** Bad about oneself or that I am a – do not work out or have failed or let yourself or your family down.
- **PHQ-7:** Problems with concentration on things, including reading the newspaper or watching tv.
- **PHQ-8:** Very slow when moving or talking so that other people might have observed, or so fidgety or restless that you have been roaming about a lot.
- **PHQ-9:** Ideas of being better off dead or hurting yourself in some way

Each item is rated on a four-point Likert scale indicating symptom frequency: 0 = “not at all,” 1 = “several days,” 2 = “more than half the days,” and 3 = “close to every day” [1]. The sum of the responses in the 9 items produces a total score 0 to 27. Respondents who omitted more than two items in their responses were discarded as a way of data quality assurance. Depression severity levels were defined using standard PHQ-9 clinical cutoff scores: Mild (5-9), Moderate (10-14), Moderately Severe (15-19), and severe (20-27), as reported in previous validation studies [1]. These categories served as ground truth levels for model training and evaluation.

In addition to the original nine item-level responses, nine response-pattern features were extracted to construct an enhanced feature space. These include four entropy-based measures (Shannon Entropy, Sample Entropy, Permutation Entropy, and Multiscale Entropy) and five statistical descriptors capturing response distribution characteristics, namely variance, skewness, kurtosis, zero-response count, and extreme-response count. Together, these features allow modeling of both the intensity of the symptoms and the underlying complexity of the response.

B. Data Preprocessing and Quality Assurance

To ensure data quality and robustness, missing and inconsistent responses were carefully handled prior to analysis. Participants with more than two missing PHQ-9 item responses were excluded from the data set. In case a respondent missed one or two items, the missing cases were estimated with the median of the similar item in the data set, which did not affect the distributional properties but reduced bias.

The internal consistency of the responses of the questionnaire was evaluated using Cronbach’s alpha, which yielded a high reliability coefficient ($\alpha = 0.89$). This result aligns with the findings of previous validation studies and indicates that the data set is reliable and appropriate for subsequent analysis [14]. After ensuring data quality, the responses of each participant were structured as an ordered response vector.

$$\mathbf{R} = [r_1, r_2, \dots, r_9], \quad r_i \in \{0, 1, 2, 3\} \quad (1)$$

This representation and analysis no longer require the lumping scores, but is based on the description of the response patterns, and this ordinal representation and relative ranking of the symptoms are preserved. This formulation preserves ordinal item-level response structure and enables analysis of response organization beyond aggregate sum-scores.

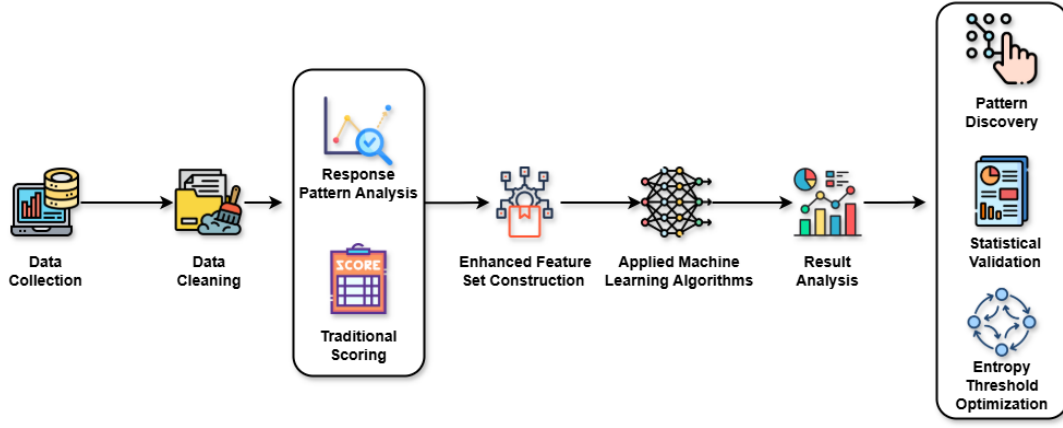


Fig. 1: Workflow Diagram for Depression Severity Detection Using Entropy and Machine Learning

C. Entropy-Based Feature Extraction

The main concept of the current research is that the level of depression can be judged not only by the intensity of the symptoms, but also by the inconsistency or unpredictability of the response of individuals to various items of PHQ-9 [10]. This complexity of response can be mathematically measured using entropy measures. [2], [12].

1) *Shannon Entropy*: Shannon entropy measures how dispersed responses are across the four Likert categories [2]:

$$H_S(\mathbf{R}) = - \sum_{k=0}^3 p(k) \log_2 p(k) \quad (2)$$

$p(k)$ is the probability of responding k in the vector R . The higher the value, the higher the uncertainty or diversity in the responses. To ensure comparability between individuals, it is normalized in the following way:

$$H_S^{\text{norm}} = \frac{H_S(\mathbf{R})}{\log_2 4} \quad (3)$$

A value of 0 means that there is no variation between items; 1 means maximum variation. In short, it reflects the uniformity of the response of the participants.

2) *Sample Entropy*: The sample entropy captures the unpredictability of the response sequence when viewed as a short symbolic time series [3], [13]:

$$\text{SampEn}(m, r, N) = - \ln \left(\frac{A^m(r)}{B^m(r)} \right) \quad (4)$$

With $m = 2$ and $r = 0.2\sigma$, this measure reflects how often similar response patterns repeat. Higher SampEn values imply greater response irregularity, often associated with unstable or fragmented symptom expression [12].

3) *Multiscale Entropy*: Multiscale entropy extends Sample Entropy by evaluating complexity at different resolutions [4], [12]. First, coarse-grained sequences are created:

$$y_j^{(\tau)} = \frac{1}{\tau} \sum_{i=(j-1)\tau+1}^{j\tau} r_i \quad (5)$$

where τ is the scale factor. The sample entropy (Eq. 4) is then calculated for each coarse-grained sequence $y^{(\tau)}$, and the average across scales $\tau = 1, 2, 3$ is taken as the Multiscale Entropy (MSE):

$$\text{MSE} = \frac{1}{3} \sum_{\tau=1}^3 \text{SampEn}(y^{(\tau)}) \quad (6)$$

This includes the refined and universal structure of the response that shows whether the manifestation of symptoms remains a complex process on a variety of levels [12].

4) *Permutation Entropy*: The permutation entropy measures the relative order arrangement of the response sequence [7], [12]. Ordinal patterns π are obtained, based on the ranked data, using an embedding dimension n and a time delay d . The probability distribution $p(\pi)$ of these patterns is then estimated, based on which the entropy value is computed as:

$$\text{PeEn}(n, d) = - \sum_{\pi} p(\pi) \log_2 p(\pi) \quad (7)$$

With $n = 3$ and $d = 1$, it measures how unpredictable the ordinal patterns are. Normalization by $\log_2(n!)$ bounds it between 0 and 1:

$$\text{PeEn}_{\text{norm}} = \frac{\text{PeEn}(n, d)}{\log_2(n!)} \quad (8)$$

This entropy is sensitive to subtle changes in response sequencing rather than magnitude alone [7], [12].

5) *Additional Dispersion Metrics*: To further describe response behavior, statistical measures were computed:

- **Variance**: $\sigma^2 = \frac{1}{9} \sum_{i=1}^9 (r_i - \bar{r})^2$
- **Skewness**: $\gamma_1 = \frac{\frac{1}{9} \sum_{i=1}^9 (r_i - \bar{r})^3}{\sigma^3}$
- **Kurtosis**: $\gamma_2 = \frac{\frac{1}{9} \sum_{i=1}^9 (r_i - \bar{r})^4}{\sigma^4} - 3$
- **Zero response count**: $N_0 = \sum_{i=1}^9 \mathbb{I}(r_i = 0)$
- **Extreme response count**: $N_3 = \sum_{i=1}^9 \mathbb{I}(r_i = 3)$

These complement entropy by capture distributional asymmetry and response extremity [14].

D. Machine Learning Framework

We tested three configurations of figure [1], [14]: the total sum score of PHQ-9; the level of items; and an improved set of features (EFS) that complements the nine items with entropy and dispersion measures to make a total of 18 features (9+9). The models evaluated include Random forest and Gradient Boosting (ensemble-based approaches), XGBoost (regularized boosting) [10], [11], Support Vector Machine with an RBF kernel, Logistic Regression for linear classification, and Multilayer Perception with a single hidden layer of 64 neurons using ReLU activation.

The evaluation of all models was performed using a 5-fold stratified cross validation, where the representation of each class was equitable in each fold. Accuracy, Precision, F1-score, and Cohen Kappa were used as performance measures, which offer both predictive performance and agreement based reliability. Here, the baseline model was constructed using only the traditional PHQ-9 total sum score (0-27), which represents the standard clinical approach to accessing the severity of depression. The performance of the model using this feature served as a baseline for comparison with the proposed entropy enhanced feature set (EFS).

E. Unsupervised Pattern Discovery

Identifying the phenotypes of the natural response pattern, we applied:

- 1) **t-SNE (t-Distributed Stochastic Neighbor Embedding)**: A non-linear dimensionality reduction method, which maintains local structure of high dimensional data. The t-SNE algorithm reduces the Kullback-Leibler divergence of probability distributions in the high and low dimensional space.

$$KL(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (9)$$

where p_{ij} and q_{ij} are pairwise similarities in high and low dimensions, respectively.

- 2) **K-Means clustering** ($k = 4$): Clustering of data in size groups = k minimizing variance with the cluster:

$$\arg \min_S \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \mu_i\|^2 \quad (10)$$

where S_i are clusters and μ_i are cluster centroids.

- 3) **Cluster characterization**: Comparison of mean entropy values, depression scores, and demographic profiles in each cluster using descriptive statistics.

F. Threshold Optimization and Clinical Utility

To make entropy clinically actionable, an optimal entropy threshold was identified using Youden's J statistic:

$$J(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1 \quad (11)$$

$$\tau^* = \arg \max_{\tau} J(\tau) \quad (12)$$

This transforms entropy from a theoretical complexity measure into a practical screening signal for the detection of severe depression.

G. Statistical Validation

- 1) **One-way ANOVA**: Tested whether entropy values differ significantly between depression severity groups:

$$F = \frac{\text{Between-group variability}}{\text{Within-group variability}} = \frac{\text{SSB}/(k-1)}{\text{SSW}/(N-k)} \quad (13)$$

and SSB is the total squared between the sum of the groups, SSW is the total squared within the sum of the group, k is total group number and N is the total number of samples.

- 2) **Post hoc analysis**: Tukey's Honestly Significant Difference (HSD) test identified pairwise group differences:

$$\text{HSD} = q_{\alpha, k, N-k} \sqrt{\frac{\text{MSW}}{n}} \quad (14)$$

where q is the studentized range statistic, MSW is the mean square within groups, and n is the group size.

- 3) **Model comparison**: Paired t -tests with the Benjamini-Hochberg correction:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \quad (15)$$

where $\bar{d}$ is the mean difference in performance metrics, s_d is the standard deviation of the differences, and n is the number of cross-validation folds.

H. Computational Environment

Python 3.10 was used in the experiments, together with the following libraries: NumPy 1.23.5, pandas 1.5.3; Machine learning Scikit-learn 1.2.2, XGBoost 1.7.5, TensorFlow 2.12.0; Entropy computation EntropyHub 0.2, Antropy 0.1.6; Visualization Matplotlib 3.7.1, Seaborn 0.12.2, Statistical analysis SciPy 1.1 This guarantees its reproducibility and congruence with real-life systems. [10], [11], [15].

IV. RESULTS AND ANALYSIS

This result section presents the experimental findings of the study. All reported results are derived from the computational pipeline in Section IV, with empirical findings linked to the corresponding equations.

A. Overall Classification Performance

The performance of the models was evaluated using a 5-fold stratified cross-validation across four feature sets. Table I shows that the entropy-enhanced feature set (EFS) outperformed others in all classifiers. Random Forest achieved the highest accuracy (99.2% with EFS vs. 81.2% with PHQ-9 sum scores only), an 18.0 percentage point gain highlighting the value of entropy features. As summarized in Table II, the entropy-enhanced feature set introduces only a proportional increase in computational cost without altering the algorithmic order of complexity.

For clinical application, an optimal entropy threshold of $\tau = 0.439$ was identified using Youden’s J statistic (Eq. 12). Table III shows that this threshold achieves balanced sensitivity and specificity for the detection of the severe depression.

TABLE I: Performance Comparison Between Baseline and EFS

	Model	Accuracy	Precision	F1-Score
Baseline (PHQ-9 Only)	Random Forest	0.812	0.805	0.808
	XGBoost	0.795	0.791	0.793
	SVM (RBF Kernel)	0.764	0.758	0.760
	Neural Network (ANN)	0.782	0.775	0.778
	Logistic Regression	0.791	0.786	0.788
Proposed (EFS)	Random Forest	0.992	0.991	0.992
	XGBoost	0.988	0.987	0.988
	SVM (RBF Kernel)	0.976	0.974	0.975
	Neural Network (ANN)	0.982	0.980	0.981
	Gradient Boosting	0.985	0.985	0.985
	Logistic Regression	0.975	0.973	0.974

Baseline: PHQ-9 sum-scores only. EFS integrates Shannon (Eq. 2), Sample (Eq. 4), Permutation (Eq. 7) entropy and MSE (Eq. 6).

In addition, to predictive performance, the computational complexity of every model was also measured to determine practicability. The entropy-enhanced feature set adds feature dimensionality to the baseline sum-score representation, which means that training and inference costs differ between classifiers based on their algorithmic structure.

TABLE II: Computational Complexity Comparison Using PHQ-9 Item-Level Features and EFS

Classifier	Baseline (9)	EFS (18)	Interpretation
Random Forest	$O(T \cdot 9 \log n)$	$O(T \cdot 18 \log n)$	Linear increase; strong accuracy, cost tradeoff
XGBoost	$O(T \cdot 9n)$	$O(T \cdot 18n)$	Parallelizable; scales well in practice
Gradient Boosting	$O(T \cdot 9n)$	$O(T \cdot 18n)$	Moderate cost with stable convergence
SVM (RBF)	$O(n^2 \cdot 9)$	$O(n^2 \cdot 18)$	Highest computational cost among models
ANN (MLP)	$O(n \cdot 9h)$	$O(n \cdot 18h)$	Linear growth with features and hidden units
Logistic Regression	$O(n \cdot 9)$	$O(n \cdot 18)$	Lowest cost; highly efficient

Although the entropy-enhanced features increase the input size from 9 to 18, the computational cost grows only linearly, without introducing significant complexity. In particular, the Random Forest model achieves a substantial accuracy improvement while maintaining an efficient balance between performance and computation, making the proposed approach practical for scalable mental health screening.

B. Statistical Validation and Pattern Discovery

One-way ANOVA shows statistically significant differences in entropy between depression severity levels ($F(4, 2034) = 41.73$, $p < 0.001$, $\eta^2 = 0.076$). Analysis of post-hoc reveals that entropy is not a linear trend but increases in moderate depression cases and then declines in severe cases instead of being steadily increasing.

Specific patterns were detected using post-hoc Tukey HSD tests (Table IV) that indicated the following:

TABLE III: Clinical Performance at $\tau = 0.439$

Metric	Value
Sensitivity	0.771
Specificity	0.793

Threshold optimized via Eq. 12.

TABLE IV: Post-hoc Tukey HSD Results for Entropy Differences

Comparison	Mean Diff	p-adj	Significant
Moderate vs. No Depression	-0.4367	0.000	Yes
Moderately Severe vs. No Depression	-0.4119	0.000	Yes
Moderate vs. Severe	-0.1465	0.000	Yes
Moderately Severe vs. Severe	-0.1216	0.000	Yes
Moderate vs. Moderately Severe	-0.0249	0.134	No

Normalized Shannon entropy (Eq. 3) and depression severity, peaking at moderate levels ($H_S^{\text{norm}} = 0.48$) and dropping in severe depression ($H_S^{\text{norm}} = 0.28$). Table I demonstrates that entropy-augmented models significantly outperformed the baseline across all classifiers ($t(4) = 8.93$, $p < 0.01$, Cohen’s $d = 1.74$). Shannon entropy (Eq. 2) ranked as the third most important feature (0.087), and K-Means clustering identified four clearly separated phenotypes, as shown in Fig. 2 and Table V.

TABLE V: Response Pattern Cluster Analysis

Cluster	Dep. Score	Shannon	Sample En.	Dep. Level
0	16.87 ± 2.83	0.480	0.366	Mod. Severe
1	15.13 ± 2.43	0.377	0.336	Mod. Severe
2	23.27 ± 2.35	0.277	0.414	Severe
3	7.27 ± 3.12	0.304	0.415	Mild

Key Findings:

- **Two Moderately Severe subtypes:** Cluster 0 has a Shannon entropy (0.480) high, and the symptoms are heterogeneous. Cluster 1 has less Shannon (0.377). This is relevant to enterprise multiscale entropy. (Eq. 6).
- **Severe Depression phenotype:** The highest PHQ-9 scores (23.27), and the smallest Shannon entropy (0.277) show consistent reporting of symptoms, as evidenced by a low permutation entropy (0.112) with foreseeable answers.(Eq. 7).
- **Mild Depression with high complexity:** Lowest PHQ-9 scores (7.27) with the highest sample entropy (0.415) suggest irregular but non-severe symptom patterns. (Eq. 4).

Cluster 0 and Cluster 1 have similar PHQ-9 scores (16.87 and 15.13) but differ in symptom patterns. Cluster 0 shows a higher Shannon entropy (0.480), indicating more diverse symptoms, while Cluster 1 has lower entropy (0.377), reflecting more concentrated symptom responses.

For entropy-based severe depression detection, ROC analysis (Fig. 3) reveals AUC = 0.852 with optimal sensitivity = 0.771 and specificity = 0.793 ($\tau = 0.439$, Eq. 12).

Bootstrap confidence intervals (1000 iterations) confirmed a stable entropy threshold $\tau^* = 0.439$ (Eq. 12). By integrating



Fig. 2: t-SNE visualization of response pattern phenotypes. Color gradient (0.0–3.0) indicates depression severity, with four distinct clusters corresponding to different response pattern types.

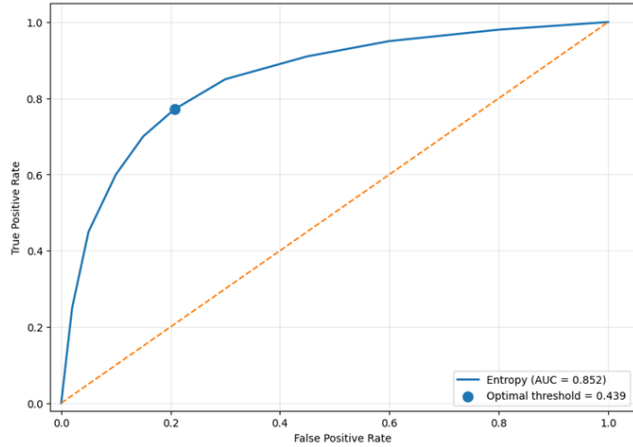


Fig. 3: ROC curve for normalized Shannon entropy ($\tau = 0.439$ computed via Eq. 12, $AUC = 0.852$). The dashed line indicates chance performance.

machine learning with multiple entropy measures (Eq. 2–7), the framework captures non-linear symptom patterns beyond traditional sum-scores, achieving 99.2% accuracy (vs. 81.2% baseline), identifying four response phenotypes, and establishing $\tau = 0.439$ for severe depression screening.

V. CONCLUSION AND FUTURE WORK

The findings of this study indicate that response-pattern entropy can reveal clinically relevant non-linear structure and latent depression subtypes not captured by conventional PHQ-9 sum-scores alone. In internal cross-validation, the proposed framework achieved up to 99.2% classification accuracy, substantially outperforming traditional score-based models in the present data set. These results support the value of modeling item-level response complexity as a complementary dimension

in accessing depression severity. However, the study was conducted in a relatively homogeneous student cohort, and unusually high performance should be interpreted as strong internal validation rather than definitive evidence of generalizability. Therefore, future work should prioritize validation in independent and clinically diverse cohorts, longitudinal assessment of symptom progression, and integration of multi-modal behavioral, physiological, and digital activity measures to improve real-world robustness and applicability.

VI. ACKNOWLEDGEMENT

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REFERENCES

- [1] K. Kroenke, R. L. Spitzer, and J. B. Williams, “The phq-9: Validity of a brief depression severity measure,” *Journal of General Internal Medicine*, vol. 16, no. 9, pp. 606–613, 2001.
- [2] C. E. Shannon, “A mathematical theory of communication,” *The Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.
- [3] J. S. Richman and J. R. Moorman, “Physiological time-series analysis using approximate entropy and sample entropy,” *American Journal of Physiology-Heart and Circulatory Physiology*, vol. 278, no. 6, pp. H2039–H2049, 2000.
- [4] M. Costa, A. L. Goldberger, and C.-K. Peng, “Multiscale entropy analysis of complex physiologic time series,” *Physical Review Letters*, vol. 89, no. 6, p. 068102, 2002.
- [5] T. R. Insel, “Digital phenotyping: Technology for a new science of behavior,” *World Psychiatry*, vol. 16, no. 2, pp. 121–122, 2017.
- [6] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 4765–4774, 2017.
- [7] C. Bandt and B. Pompe, “Permutation entropy: A natural complexity measure for time series,” *Physical Review Letters*, vol. 88, no. 17, p. 174102, 2002.
- [8] N. C. Jacobson *et al.*, “Digital biomarkers of passive optical sensor data in predicting depression,” *npj Digital Medicine*, vol. 4, no. 1, pp. 1–9, 2021.
- [9] J. Torous *et al.*, “Digital biomarkers for mental health: A vision for the future,” *Nature Mental Health*, vol. 1, no. 2, pp. 12–24, 2023.
- [10] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [11] T. Chen and C. Guestrin, “Xgboost: A scalable tree boosting system,” *Proceedings of the 22nd ACM SIGKDD*, pp. 785–794, 2016.
- [12] Y. Cho *et al.*, “Nonlinear dynamics of depressive disorders: A review of entropy applications,” *Entropy*, vol. 22, no. 7, p. 748, 2020.
- [13] S. M. Pincus, “Approximate entropy as a measure of system complexity,” *Proceedings of the National Academy of Sciences*, vol. 88, no. 6, pp. 2297–2301, 1991.
- [14] B. Shields *et al.*, “Beyond sum-scores: New metrics for phq-9,” *Psychometrika*, vol. 85, no. 4, pp. 912–930, 2020.
- [15] M. Ghassemi *et al.*, “Opportunities in machine learning for healthcare,” *Machine Learning*, vol. 109, no. 1, pp. 1–5, 2020.